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SILICON CARBIDE MOSFET TECHNOLOGY†

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Abstract—The research and development activities carried out to demonstrate the status of MOS planar technology for the manufacture of high temperature SiC ICs will be described. These activities resulted in the design, fabrication and demonstration of the world's first SiC analog IC—a monolithic MOSFET operational amplifier. Research tasks required for the development of a planar SiC MOSFET IC technology included: characterization of the SiC–SiO_2 interface using thermally grown oxides; high temperature (350°C) reliability studies of thermally grown oxides; ion implantation studies of donor (N) and acceptor (B) dopants to form junction diodes; epitaxial layer characterization; device isolation methods; and finally integrated circuit design, fabrication and testing of the world's first monolithic SiC operational amplifier IC. High temperature circuit drift instabilities at 350°C were characterized. These studies defined an SiC depletion model MOSFET IC technology and outlined tasks required to improve all types of SiC devices. Copyright © 1996 Elsevier Science Ltd

1. INTRODUCTION

Although SiC with its intrinsic advantages (Table 1) will undoubtedly replace Si and GaAs devices in some applications and make new applications possible, it will not supplant the volume of Si devices used today where low cost, functional density and very moderate operating temperatures are the primary driving forces.

The fact that SiO_2 has the highest dielectric strength of all insulators and thin films of this material can be thermally grown on Si in dry oxygen or steam ambient was a key factor in making Si MOSFETs and MOS ICs possible. Neither Ge (Si's predecessor) or the compound semiconductor GaAs enjoys this advantage. However, SiC does have this capability. The purpose of this paper therefore is to describe the status of SiC MOSFETs and SiC MOSFET ICs and to see if SiC MOSFET technology might follow a similar technological path defined by Si MOSFET's past history. The understanding of this status was garnered during the course of an effort to develop an SiC MOSFET IC technology for fabricating high temperature control chips for distributed sensor/control systems. Previous work defined a mesa SiC UV photodiode for high temperature optical sensor applications[1] which led to the development and production of SiC flame sensors for the control of large gas turbines[2].‡ These applications are made possible by SiC's large band gap of approximately 3 eV.

In the present work, the NMOS depletion mode IC technology that was developed avoided several technical problems that were identified. These were: use of Al doped *p*-type SiC produces highly variable *n*-channel inversion mode NMOS FETs of poor quality because of large amounts of oxide fixed-charge and variable fast and slow interface state trap densities; high temperature gate oxide reliability is poor with positive gate bias and the quality of thermal oxides grown on N ion-implanted diode regions is considerably degraded. These studies and the resulting SiC operational amplifier IC[3] are described in the following sections.

SiC material

SiC boules are grown using the seeded modified Lely technique that has been extensively described elsewhere[4–7] and research is done using SiC epitaxial layers grown on substrates cut from these boules. These layers are formed using CVD. Methods of doing this have also been described[8–10]. The most readily available epitaxial layers are either N or Al doped, but recently B-doped layers have also been produced[11]. Whereas there are a large number of SiC polytypes with differing bandgaps and carrier mobilities, most work has been done using the 6H SiC crystal. Mobility values as a function of doping concentration are uncertain. Nevertheless, calculations for 6H SiC were carried out using the best information available on carrier mobility and activation energies of the intentionally added donor and acceptor impurities (Table 2). The results of these calculations have been published elsewhere[12].

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Table 1. Comparison of the properties of SiC, Si and GaA

Semiconductor	6H-SiC	Si	GaAs
Bandgap (eV at 27°C)	2.90	1.12	1.43
Thermal conductivity ($W\ cm^{-1}\ °C^{-1}$ at 27°C)	5.0	1.5	0.5
Saturated electron drift velocity ($cm\ s^{-1}$)†	2×10^7	1×10^7	5×10^6
Electron mobility ($cm^2\ V^{-1}\ s^{-1}$ at 27°C)	250	1400	8500
Hole mobility ($cm^2\ V^{-1}\ s^{-1}$ at 27°C)	50	600	400
Breakdown electric field ($V\ cm^{-1}$)	40×10^5	3×10^5	4×10^5
Dielectric constant (dimensionless)	10.0	11.8	12.8

†At $E \geqslant 2 \times 10^5\ V\ cm^{-1}$.

Epitaxial layers

One inch diameter Si-faced 6H SiC wafers with *n*- and/or *p*-type epitaxial layers were purchased from Cree Research.† Epitaxial layers have also been recently obtained from the NASA group in Cleveland.‡ Both heavily doped *n*- and *p*-type substrates were utilized. For IC fabrication the doping and thicknesses of the epitaxial layers were specified for obtaining the proper threshold and back-gate bias constant for MOSFET integrated circuits. Expitaxial layer doping profiles were obtained using a dual point Hg Schottky barrier probe and MOS capacitors[13]. Top sided multiple capacitor MOS measurements were utilized, to avoid the need for a backside contact. These measurements, combined with data on completed MOSFETs and ICs, showed that control of the doping and thickness of the epitaxial layers were the most significant factors in controlling yield and reproducibility. Errors in doping concentration measurement techniques were a contributing factor to this doping control problem.

THIN THERMAL GATE OXIDE STUDIES

For MOS devices the characteristics and understanding of gate oxides are vital. Studies were carried out in five areas: (a) thermal oxidation; (b) dielectric strength; (c) $SiC–SiO_2$ interface characteristics; (d) time-dependent dielectric breakdown (TDDB); and (e) correlation of items (b–d), with MOSFET device data.

Thermal oxidation of SiC (experimental methods)

All the samples used Si-faced 1 in diameter SiC wafers with a 3 μm epitaxial layer grown on heavily doped substrates. For the *n*-type wafers the epitaxial layer was N doped to a level of between 3 and $5 \times 10^{15}\ cm^{-3}$ on a $1–2 \times 10^{18}\ cm^{-3}$ N-doped substrate. The *p*-type epitaxial layer was Al doped at a concentration of between about 1 and $2 \times 10^{16}\ cm^{-3}$ on $1–10 \times 10^{17}\ cm^{-3}$ Al-doped substrate. The wafers were cleaned using a hot mixture of H_2SO_4 and H_2O_2 for 15 min followed by 10 s in 10% HF. The wafers were oxidized in steam at 1 atm pressure at 1100°C for 380 min to form 0.05 μm of SiO_2. This was followed by an N_2 anneal for 1 h at 1100°C. A Mo film about 0.5 μm thick was deposited using magnetron sputtering and subsequently patterned into MOS capacitor dots. The thickness of the SiO_2 layer was measured using an ellipsometer and verified by MOS capacitance measurements. A number of alternative oxidation methods were also utilized with no effect on the reliability results described below.

Table 2. Donor and acceptor activation energies for 6H SiC

Impurity	Activation energy	Concentration
Nitrogen donor	Ec − 0.078 eV	$4 \times 10^{18}\ cm^{-3}$
	Ec − 0.1 eV	$1–2 \times 10^{16}\ cm^{-3}$
Aluminum acceptor	Ev + 0.2 eV	$0.8–2 \times 10^{17}\ cm^{-3}$
Boron acceptor	Ev + 0.3 eV	$5 \times 10^{13}\ cm^{-3}$

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Dielectric strength (virgin samples)

The dielectric strength and defect density of thin 500 Å thermally grown oxides on SiC wafers are comparable to those grown on Si wafers, with an absence of low voltage breakdown defects. The dielectric strength of SiO_2 is about $10\ MV\ cm^{-1}$, so most of the breakdowns are intrinsic in nature as shown in Fig. 1(A).

Dielectric strength (ion-implanted samples)

Because the gate of an enhancement or normally off MOSFET must overlap the source and drain diodes initially formed by implantation before gate formation in a nonself-aligned gate process, it was important to determine the dielectric strength and/or breakdown defect density of oxides grown on ion-implanted regions. N "box profile" implantations utilized to make the source and drain diodes of SiC *n*-channel MOSFETs were used to make the samples for these studies. Dielectric breakdown events occurred between 1 and $6\ MV\ cm^{-1}$ for the thermal oxides grown on these surfaces [Fig. 1(B)]. Annealing the ion-implanted samples at temperatures as high as 1500°C before oxidation did not decrease the frequency of low field breakdown events.

Time-dependent dielectric breakdown (TDDB) mean-time to failure (MTF)[3]

Time-dependent dielectric breakdown is a well-known wear-out mechanism for Si MOS devices. However, no TDDB data had been generated for SiC MOS devices. Published data for Si MOS devices was

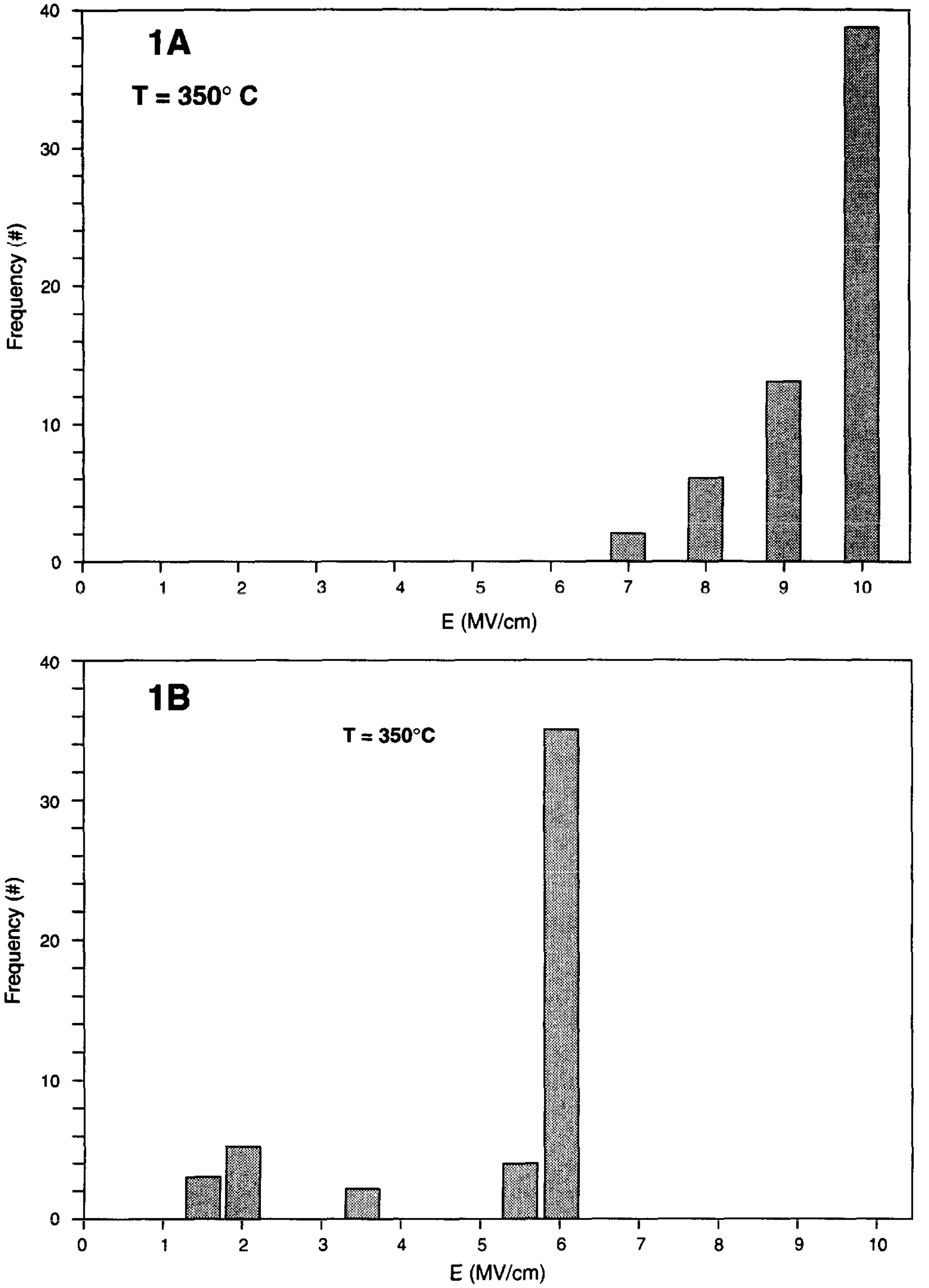

Fig. 1. 350°C dielectric breakdown histogram of 500 Å of thermally grown oxide SiC MOS capacitors. Virgin wafer (A) and wafer N ion-implanted before oxidation (B).

limited to temperatures no higher than 250°C and showed that the MTF was very short[14].

The field acceleration factor, B, is commonly used to characterize SiO_2 TDDB data since MTF is shortened by 10^{-B} with an increase in field strength of 1 MV cm^{-1}. Large B values indicate long MTF. All measurements were taken at 350°C utilizing a self-tending automatic probe station and a computerized data collection and control system. Table 3 and Fig. 2 summarizes the results giving B and extrapolated MTF at 2 MV cm^{-1} which is the nominal operating field for a MOSFET. The MTF with positive bias applied to the gate is very short, but for negative bias applied the MTF is long. These results explained the very rapid failure rate of SiC n-channel enhancement mode devices at high temperatures. Electron injection into the oxide from the SiC substrate degrades the oxide with time and

Table 3. SiC MOS gate dielectric reliability characterization, time-dependent breakdown analysis at 350°C

Gate oxide type	Field acceleration factor (β)	MTTF at $E = 2$ MV cm^{-1} (days)
	n-SiC (6H) + gate bias (virgin samples)	
0.05μ/1100C steam	0.5 cm MV^{-1}	1×10^{-1}
0.09μ/1000C steam	0.6 cm MV^{-1}	2×10^{0}
0.05μ/1200C dry	0.4 cm MV^{-1}	1×10^{-2}
	p-SiC (6H) − gate bias	
Gate oxide type	Field acceleration factor (β)	MTTF at $E = 2$ MV cm^{-1} (years)
0.05μ/1100C steam	2.1 cm MV^{-1}	4×10^{4}
0.05μ/1100C steam	1.7 cm MV^{-1}	1×10^{6}
0.05μ/1200C dry	1.5 cm MV^{-1}	3×10^{5}
	n-SiC (6H) + gate bias (N ion-implanted samples)	
Gate oxide type	Field acceleration factor (β)	MTTF at $E = 2$ MV cm^{-1} (days)
0.05μ/1100C steam	0.1–0.5	$< 5 \times 10^{-1}$

leads to rapid failure rates when a positive bias is applied to the gate electrode. This is most probably due to the fact that SiC surfaces are microscopically very rough and irregular[15]. Another factor could be the presence of metallic impurities in the SiC wafer that can be incorporated in the thermally grown oxide.

The strong dependence on polarity of MTF that was observed on SiC MOS samples should not be construed to be an intrinsic property of SiO_2 layers on semiconductor substrates. Studies at the National Institute of Standards and Technology (NIST) prompted by the above results and carried out at temperatures between 300 and 400°C show that Si MOS samples can have very long MTF for both positive and negative gate bias[16,17].

SiC MOS INTERFACE CHARACTERISTICS[18]

A number of conclusions can be obtained from a detailed study of MOS C(V) curves. The lateral shift in voltage of the experimental curve along the voltage axis can be utilized to determine the fixed-space change in the oxide. An estimation of the density of fast interface states in the bandgap can be obtained using the Terman method[19].

For MOS C(V) curves using Al-doped p-type substrates, the results can be summarized as follows. The oxide space charge is positive and high at about 1.5×10^{12} cm^{-2}. In addition, C(V) curve hysteresis indicates that these oxides contain a large number of slow traps. Reduction of the MOS C(V) curves to obtain the fast interface state density as a function of surface potential shows that the total number of fast interface states is large, being about equal to 3×10^{12} cm^{-2} and distributed in the bandgap near the valence band edge (Fig. 3). This is not sufficient to prove that there is a large number of fast interface states near the conduction band edge, but it indirectly supported the enhancement mode NMOS FET effective channel mobility data described in a later section.

Surprisingly, quite the opposite was found using n-type wafers[18]. The experimental curve is compared to a theoretical curve using the equations previously published[20]. A work function difference based on the Mo work function (4.69 eV) and an electron affinity of SiC (4.21 eV) was employed to shift the theoretical curve 0.5 V. However, no arbitrary fitting parameters were used. These results show that the C(V) curves at room temperature are "ideal" (Fig. 4) with no hysteresis (absence of slow traps), very low offset voltage (very low $\leqslant 10^{10}$ cm^{-2}

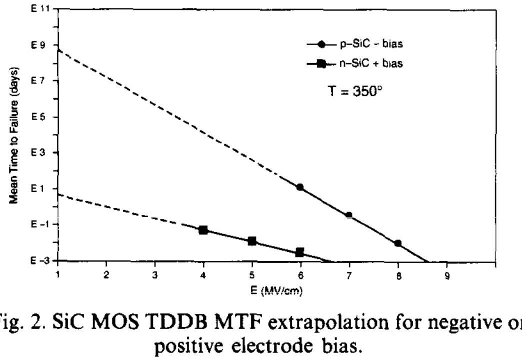

Fig. 2. SiC MOS TDDB MTF extrapolation for negative or positive electrode bias.

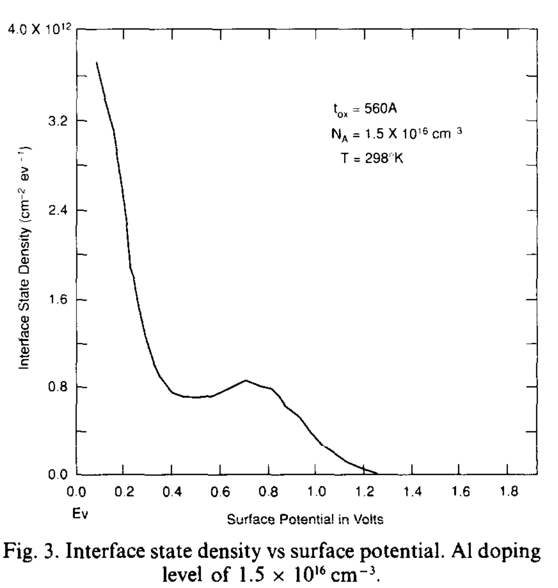

Fig. 3. Interface state density vs surface potential. Al doping level of 1.5×10^{16} cm^{-3}.

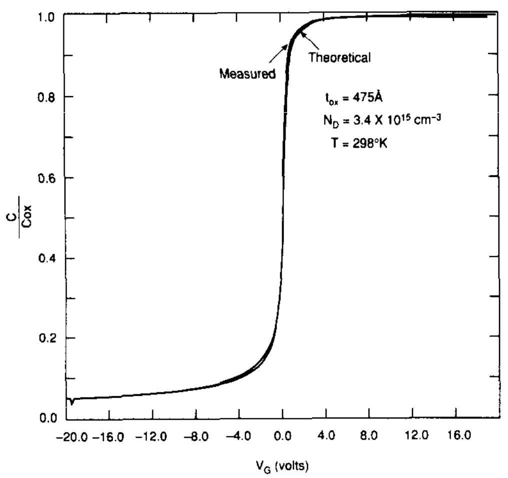

Fig. 4. Experimental and theoretical normalized capacitance vs bias curves for n-type N-doped 6H SiC MOS sample. N doping of $3.4 \times 10^{15}\,cm^{-3}$. The Mo gate work function is 4.69 eV and an electron affinity for SiC of 4.2 eV was used. There are no arbitrary fitting parameters.

oxide space charge) and absence of curve distortion (very low fast interface state density)[18]. Although these results are somewhat qualitative, they are sufficient to explain many of the fundamental issues not previously understood. These n-type wafer MOS results were very reproducible as long as the n-type epitaxial layer was grown on an N-type substrate.

The striking contrast between the MOS characteristics on p- and n-type SiC wafers indicates that the difference is probably caused by the fact that the p-type wafers are Al doped and the n-type wafers are N doped. The redistribution of impurities that occur during the thermal oxidation of SiC and Si behaves in a similar fashion. N-type dopants are rejected by the oxide during growth whereas p-type dopants are incorporated into the oxide[21]. Therefore the Al dopant in the oxide is likely to be the cause of the p-type SiO_2–SiC interface characteristics described above. It can be surmised that the Al incorporated into the oxide during oxidation becomes oxidized to an unknown degree producing numerous fast "dangling bond states" and slowly responding "slow" electronic trapping centers. B-acceptor impurities in contrast to Al impurities are known not to cause problems in Si MOS devices. The reason may be due to the fact that B_2O_3 is a network former whereas Al_2O_3 is a network modifier. Whether or not B doping of SiC can resolve these difficulties needs yet to be determined. Workers evaluating this possibility need to take care that any B-doped layer is not contaminated with Al.

ION IMPLANTATION

Both p- and n-type ion implantations are required for a number of device structures. To this end, ion implantation studies of N (a donor dopant) and B (an acceptor dopant) were carried out[22,23]. A summary of these results follows.

Nitrogen ion implantation[22]

Comparative experiments were carried out using implantation wafer temperatures ranging from 25 to 1000°C with subsequent annealing done at 1100–1500°C. It was found necessary to utilize a closed SiC container during the anneal cycle to prevent excessive surface pitting caused by the evaporation of Si from the wafer surface.

Because a box profile is desired for making diodes and the diffusion constants of impurities in SiC (which even at the highest temperatures are negligible), it is necessary to use multiple ion implantation energies. The doping profile of the ion-implanted region was modeled using a computer program. A resultant profile calculation together with the energies and doses of the ion implantation sequence is given in Fig. 5. Since it is difficult to reduce the energy sufficiently in order to place the peak of the concentration at the surface, 300 Å of a "screen" SiO_2 layer is utilized to reduce the range of the lowest energy implant. This is important to maximize the concentration of dopant at the surface and thus minimize the contact resistance of metal contacts applied to these regions.

The highest degree of activation was obtained by using a 1000°C ion-implantation temperature and a 1300°C annealing temperature. Diodes formed using this process have five orders of magnitude less leakage than Si diodes at 300°C. Comparison of the measured sheet resistance values with conductivity calculations indicated that essentially all or 80–100% of the implanted N atoms were electrically active.

Four terminal Kelvin contact resistance test structures were also fabricated. Data from these devices showed that the contact resistance between the sintered Ni contact and the N-doped ion implanted regions was about $10^{-5}\,\Omega\,cm^2$. Contact sintering was done for 2 min at 925°C in an Ar atmosphere.

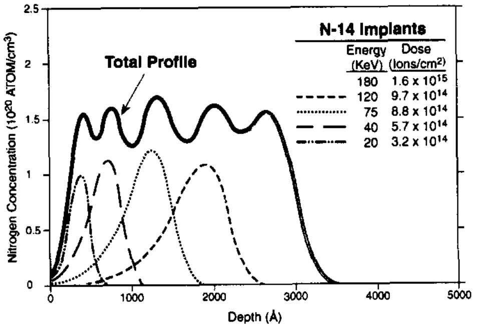

Fig. 5. Calculated profile for a sequence of nitrogen implants in SiC that were optimized by adjusting dose and energy to generate a rectangular profile.

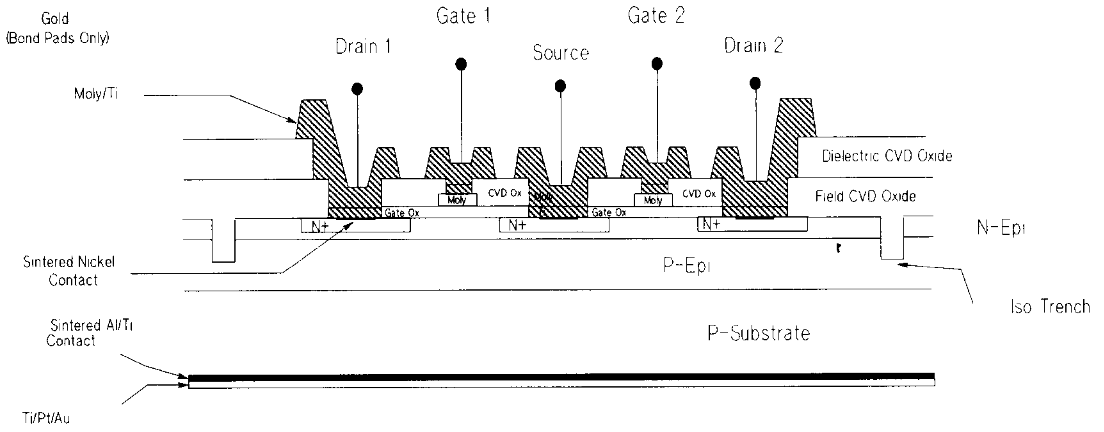

Fig. 6. SiC dual gate MOSFET cross-section.

Charge trapping in ion-implanted diodes[2, 24]

In spite of this improvement in diode leakage a further difficulty has been discovered. It has been shown that $n + p$ diodes formed by N implantation have very long recovery times after forward biasing. This is caused by carriers trapped in the junction region. Reverse current continues to flow until this space charge is neutralized by slow recombination mechanisms. This problem is not characteristic of diodes formed using n- and p-type epitaxial layers. Ion implantation produces deep level trapping centers because all the crystal damage produced by the implantation process is not annealed by post-implantation high temperature annealing steps. It is not known if higher annealing temperatures could eliminate this problem. Fortunately for the formation of n-channel depletion mode devices, n^+-contact regions formed by N implantation into n-type epitaxial layers do not exhibit charge trapping providing the ion implantation depth is shallower than any n-type layer grown over p-type SiC.

Boron ion implantation[23,25]

Diodes using B implantation were also made using n-type SiC wafers. These experiments demonstrated the successful conversion of n-type SiC to p-type, required to make very high voltage, low reverse leakage diodes. Here again the energies and implant doses were calculated to make a box profile of uniform concentration by using five successive implants. Two implantation temperatures were utilized (25 and 1000°C) to determine if diode yield would be improved with the higher temperature implant. Annealing temperatures were varied between 1000 and 1400°C.

A control sample that did not receive a B implant was also made. All samples were metallized using the Ti–Al metal contact used for contacts to p-type SiC[26]. The control diode was no more than a Schottky barrier diode with a low forward turn on voltage and high reverse leakage whereas the B implanted sample exhibits the proper junction diode characteristic of 2 V diode turn-on voltage, low reverse leakage and a high reverse breakdown that occurred at 650 V. The yield of low leakage diodes using 1000°C implantation was higher than those implanted at 25°C. More details of these experiments can be found elsewhere[23].

SiC MOSFETs

Device structure

Both n-channel enhancement (inversion) mode and depletion mode MOSFETs were fabricated and tested. Both single and dual gate devices were made. The single gate device was a simple annular gate device utilized to develop basic understanding of SiC MOSFETs and how to make them. The cross-section of the dual gate device consisting of two gates, two drains and a common source is shown schematically in Fig. 6. This device represents the differential stage of an operational amplifier. The degree of mismatch or offset voltage of this differential pair is an important parameter for most circuit applications.

The cross-section in Fig. 6 illustrates the following features of these device structures. The starting material for both n-channel enhancement and depletion mode devices could be either a p-or n-type wafer. The n-channel depletion mode device requires an added n-epitaxial channel layer as shown. Another difference is that enhancement mode devices by necessity must have a degree of overlap of the gate over the adjacent edges of the source and drain diodes. This is required to produce an n-type inversion surface channel between the source and drain diodes by the application of positive gate bias. The n-channel depletion mode device is normally on and requires negative bias to turn off the channel. Therefore (as shown) this latter device does not require gate overlap of the source and drain diodes. These differences have obvious device reliability implications because of the previously described results on oxide defects and reliability.

Additional details shown in Fig. 6 are: the N ion-implanted regions; 500 Å of thermally grown gate oxide; Mo gate electrodes; CVD field oxide; sintered Ni contact to source and drain diodes; phosphorous doped CVD glass passivation layer; Mo–Ti interconnect metal and the backside Al–Ti; and Ti–Pt–Au metallization utilized for die bonding the chips to Au-plated package headers. Also, but not shown, is the Au layer on the bonding pads. Au wires were utilized to connect the bonding pads to Au plated pins in ceramic packages. Further information on the use of refractory metals in semiconductor devices has been published[27–33].

N-channel enhancement mode devices

A number of wafers were processed early in the study when the objective was to determine if n-channel enhancement mode devices were feasible for integrated circuit components. The device structure and process flow were identical to that utilized to make n-channel depletion mode devices except for the aforementioned differences. The p-type epitaxial layer doping concentration was either between 2 and $3 \times 10^{16}\,cm^{-3}$ or about $2 \times 10^{15}\,cm^{-3}$.

SiC NMOS enhancement or inversion mode devices built on p-type substrates have been shown to have good characteristics at high temperatures[34,35], but the transconductance as a function of temperature has not behaved according to theory. Surprisingly these previous works neglected to extract the mobility from the transconductance data. The NMOS problem makes itself evident when device conductance data is reduced to a plot of effective channel mobility vs temperature for a device using a p-type epitaxial layer doped with AL at a concentration of about $2\text{–}3 \times 10^{16}\,cm^{-3}$ (Fig. 7). The effective mobility was obtained using the equation:

$$\mu = (dI/dV_g)/Cox \cdot W/L \cdot (V_g - V_t),$$

at a fixed drain voltage high enough for current saturation and where dI is the change in current produced by a small change in the gate voltage (V_g) with Cox being the gate oxide capacitance, W/L is the transistor channel width to length ratio and V_t is the threshold voltage. The term effective channel mobility is utilized because the quantity of electron

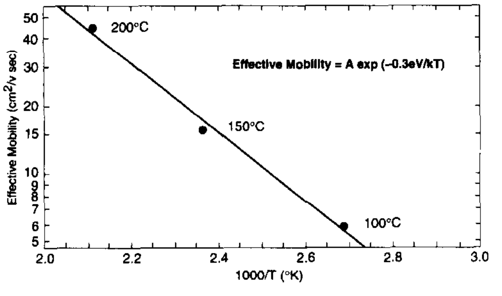

Fig. 7. n-Channel inversion mode effective channel mobility vs reciprocal temperature. ($n_A = 2\text{–}3 \times 10^{16}\,cm^{-3}$).

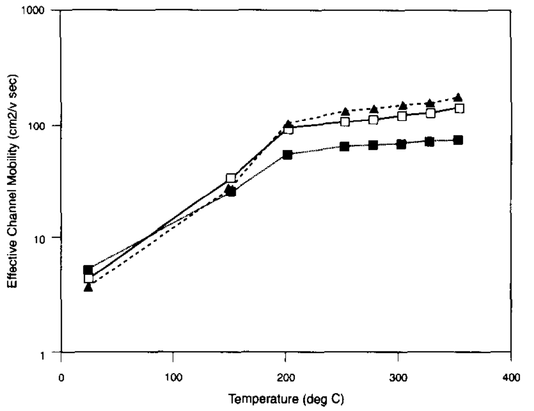

Fig. 8. Effective channel mobility of an n-channel inversion mode SiC MOSFET using a lightly doped p-type substrate. ($n_A = 2 \times 10^{15}\,cm^{-3}$).

surface charge given by the product $Cox(V_g - V_t)$ is not the conduction band free electron charge in the surface channel under the gate electrode if a fraction of the surface charge is trapped. As shown in Fig. 7, the effective channel mobility is extremely small, less than 5 below 100°C and even less at 25°C. These results are similar to values that can be obtained using previously published transconductance data[35–37]. Also, the effective channel mobility increases with increasing temperature and has an activation energy of 0.3 eV. This indicates that electrons are being thermally activated from interface state traps 0.3 eV from the conduction band edge.

The distribution of fast interface states in Si MOS devices is symmetrical with a minimum near the center of the bandgap[38]. It appears that this type of fast interface state distribution is also present at the SiC–SiO_2 interface. Notice that the "beginning" of the fast interface state density at about 0.3 eV from the valence band edge as determined from MOS $C(V)$ data (Fig. 3) agrees with the thermal activation energy of the inversion "effective channel mobility". These two observations support a hypothesis that the fast interface state distribution is similar to that of the Si–SiO_2 interface being symmetrical with a minimum near the middle of the band gap and rising rapidly as the conduction bands are approached. In addition, the effective channel mobility increases rapidly with temperature because the number of electrons in the trap states decrease as the concentration of free electrons in the surface inversion layer in the conduction band increases, according to Fermi–Dirac statistics.

Results using the more lightly Al-doped $2 \times 10^{15}\,cm^{-3}$ p-type epitaxial layer are similar, but with two notable exceptions as shown by Fig. 8. The mobility values between 25 and 100°C are higher and at 200–250°C the values rise to be between 100 and $200\,cm^2\,V^{-1}\,s^{-1}$. The presence of fast interface state traps, even with a 10-fold decrease in Al doping concentration, is still seen because of this increase in effective channel mobility with increases in temperature. These results on finished *n*-channel inversion mode MOSFETs were in the main unpredictable and irreproducible.

All these findings have implications for power devices as well as for small signal devices and low voltage ICs. For instance, in a power MOSFET or IGBT (insulated gate bipolar transistor) switch, the enhancement mode *n*-channel MOSFET needs to have a heavily doped *p*-region to eliminate punch through breakdown. This requirement reduces the MOSFET channel mobility drastically. As a result, the turn on currents will be extremely small.

n-*Channel depletion mode devices*

The understanding that was brought about by all the previously described studies indicated that *n*-channel depletion mode devices would be suitable for near-term integrated circuit fabrication. Another choice, compatible with the facts, could have been *p*-channel inversion mode, but this would result in a much lower gain bandwidth product. Both of these device types made using *n*-type epitaxial layers would be expected to behave in a normal fashion with channel mobilities that decrease with increasing temperature due to crystal lattice phonon carrier scattering.

Calculations were made to determine the effect of epi-layer doping concentration on the turnoff threshold voltage, back-gate bias voltage and surface inversion voltage. Figure 9 shows a result of some of these calculations. The lower of the two curves shows the gate turn-off threshold voltage and the upper gives the surface inversion gate voltage both plotted as a function of doping of the *n*-type epitaxial layer

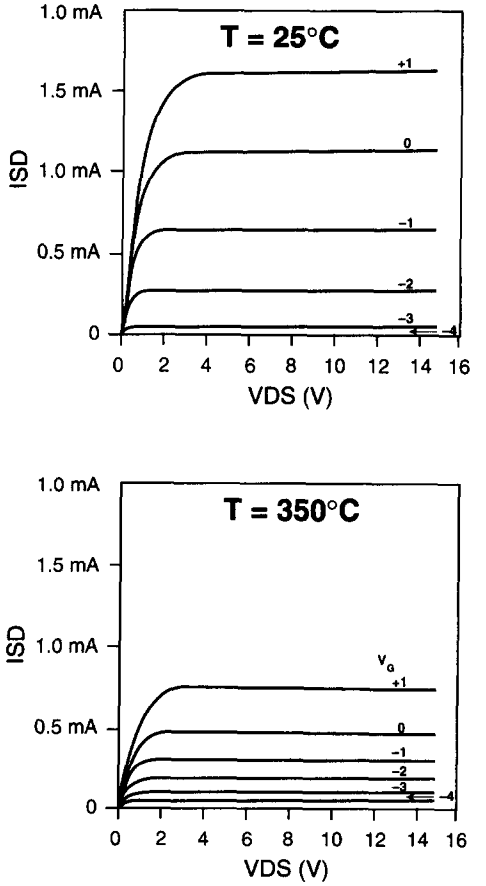

Fig. 10. ISD vs VDS of *n*-channel depletion mode SiC MOSFETs at 25 and 350°C.

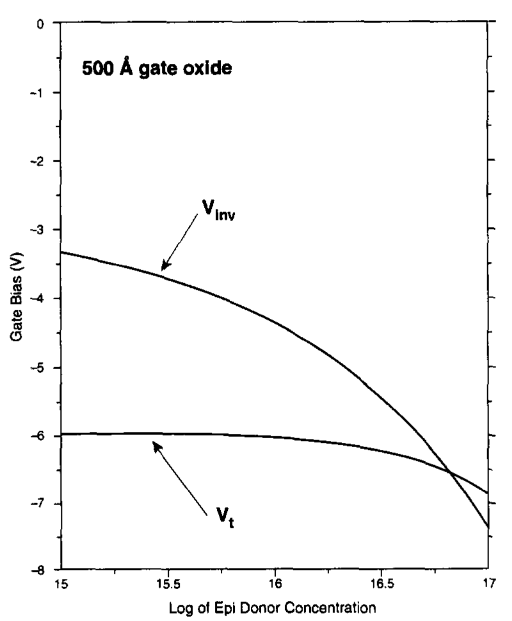

Fig. 9. *n*-Channel depletion mode gate bias curves for surface inversion and threshold voltage for channel turnoff vs 0.35 μm epitaxial layer donor concentration.

0.35 μm thick. It is important in the design of these devices that the turn-off threshold voltage be less negative than the surface inversion voltage. If the reverse were true, hole carriers would be collected at the surface and if surface inversion occurs before channel turn-off, this would result in unstable device characteristics. In the example shown in Fig. 9, the donor concentration must be about $1 \times 10^{17}\,cm^{-3}$ to eliminate this potential problem, and a thick lightly doped *p*-type epitaxial layer is required to increase the backgate bias voltage to prevent turn off of circuit nodes in an IC.

A typical set of SiC *n*-channel depletion mode device characteristics are given in Fig. 10. The electron channel mobility at 25°C was between 80 and $130\,cm^2\,V^{-1}\,s^{-1}$ and as expected decrease with increasing temperature[39]. Mobility values were extracted using the methods for MOSFET depletion mode devices described in Ref. [40]. These device characteristics were utilized in the H spice circuit model for designing and simulating the operational amplifier IC.

The offset voltage of fabricated dual gate devices was found to be highly variable. In some cases the mismatch of two adjacent gates only separated by a few microns would range between 0 and 700 mV, but in other wafers the mismatch was less than 40 mV. The cause of this variability and sometimes large offset voltage is not as yet understood. It is hoped that the use of epitaxial layers with a lower degree of compensation will reduce the variability and size of the offset. This issue is discussed next.

Al contamination and compensation in the n-*type epitaxial layer*

During the course of the extensive quality control work required to obtain wafers for device fabrication it became apparent the *n*-type epitaxial layer was contaminated with variable amounts of Al, thereby compensating this layer. The probable cause was Al evaporation from the backside of the *p*-type wafer. Device fabrication efforts then were changed to utilize *n*-type substrates. Unfortunately, the presence of the *p*-epitaxial layer still resulted in Al contamination of the *n*-type layer during its growth. This is not too surprising given the extremely high vapor pressure of Al. The presence of Al in the *n*-layer was evidenced by the amount of slow trap hysteresis in the MOS *C*(V) curves. Correlation with SIMS analytical data verified these observations. No Al was detected whenever the *n*-layer was grown on an *n*-type substrate. However, Al was detected whenever the *n*-layer was grown on a *p*-type substrate or even over a *p*- epi-layer on an *n*-type substrate. The use of site competition epitaxial growth[10]† can reduce the incorporation of Al in the *n*-layer[41].

SiC OPERATIONAL AMPLIFIER IC

Device isolation

Trench isolation of individual circuit elements was deemed to be the most practical at this point in time of SiC IC development. Since the ends of linear gate devices are required to pass over isolation trenches there was a concern that the gate oxide would be degraded in the trench. Test data showed that this should not be a problem. Nevertheless, two operational amplifier layouts were composed, one using linear gates and the other annular gate MOSFETs. With annular gates there is no gate end and in this case the device is isolated by a trench surrounding the outer junction. Thus there is no possibility of any gate leakage resulting from the isolation structure.

†In site-competition epitaxy, dopant incorporation is controlled by adjusting the Si/C ratio within the growth reactor to affect the amount of dopant atoms incorporated into substitutional SiC crystal lattice sites. These sites are either carbon-lattice sites or silicon-lattice sites located on the active growth surface of the SiC crystal. The concentration of *n*-type dopant atoms, which can occupy only C-sites, incorporated into a growing SiC epilayer can be decreased by increasing the carbon-source concentration so that C "outcompetes" N for the available C-sites. Analogously, the amount of *p*-type dopant atoms incorporated, which can only occupy Si-sites, can be decreased by increasing the silicon-source concentration within the growth reactor so that Si "outcompetes" Al for the Si-sites[10].

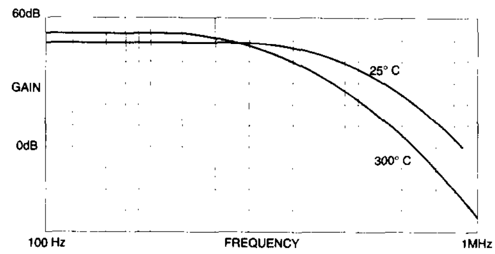

Fig. 11. Gain vs frequency at 25 and 300°C.

Operational amplifier design

Patterned SiC epitaxial resistors formed from the *n*-type epitaxial layer and isolated by the etched trench were utilized for load resistors. One advantage is that since the mobility decreases with increasing temperature the gain product of MOSFET transconductance times load resistor is constant. Circuit bias levels were set proportional to the ratio of resistances, since ratios would be more nearly temperature independent than individual resistor values.

Differential gain stages, voltage translation stages and output stages were individually simulated using an SiC MOSFET Hspice model based on the transistor's I–V parameters (Fig. 10) and a resistor model derived as a function of *np*-junction bias to account for internal circuit backgate biasing. The resistors were sized by entering the design resistance values into the Hspice simulation and the DC operating points for each node determined. The transistor operating point was set at the point of maximum transconductance per unit of current as determined by G_m/i versus V_g measurements of individual transistors. This maximizes the gain for any given operating bias voltage.

Open and closed loop gain, unity gain frequency response and step response were calculated for temperatures of 25 and 350°K. The model predicted an open loop gain of 60 dB. The chip layout was based on 7 μm dimensional layout rules.

Operational amplifier test results

A gain vs frequency plot at 25 and 300°C is given in Fig. 11. Table 4 gives the amplifier gain and bandwidth vs temperature. Similar results were obtained for both linear and annular gate structures. A photomicrograph of a die of the annular gate

Table 4. SiC operational amplifier gain and bandwidth vs temperature

T (°C)	Gain (dB)	Bandwidth (kHz)
25	49	724
160	54	501
210	54	447
250	53	380
300	53	269

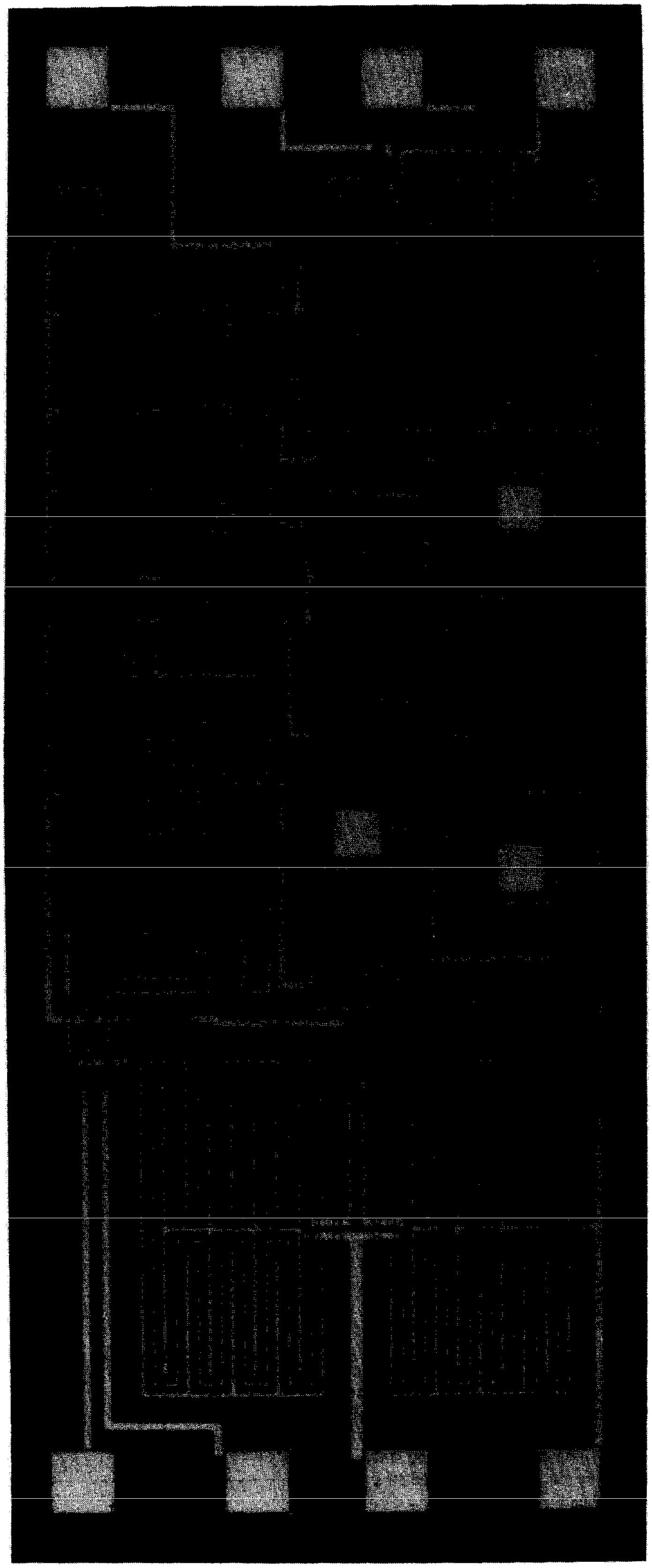

Fig. 12. Photo-micrograph of operational amplifier 1×2 mm^2 IC chip using annular gates.

design is shown in Fig. 12. A picture of the linear gate design can be found in Ref. [42].

Operational amplifier reliability

High temperature operation of packaged amplifiers was studied in detail. After brief periods of time at 350°C the circuit stopped operating. Heating the circuit without bias caused the circuit to recover, but operation at 350°C always shut down the circuit in a brief time. Tests of the behavior of individual resistors from the test element sets included in the operational amplifier wafers gave the following information.

The value of a patterned n-type expitaxial layer SiC resistor is increased in a few hours with application of negative backgate bias to the p-epitaxial layer and heating at 350°C. Subsequent heating with no bias produces recovery to the resistor's original resistance value as measured without any backgate biasing. The increase in resistance is proportional to the backgate bias voltage applied during the heat cycle. The increase can be very large (60–1000%). This correlates with the above observations on the operational stability of the amplifier. Similar instabilities have been observed for JFETs; it has been suggested that this type of instability may be due to mobile carrier charge trapping[43]. Recent results obtained from planar Schottky diodes using lightly doped n-type material indicate that trapping can occur in the surface of the SiC and produce surface depletion layers surrounding the rectifying Schottky contact[44]. 300°C stress tests with reverse bias applied to epitaxial mesa $n^{++}p$ photodiodes gave no indication that their diode characteristics and p-layer doping profile were altered. Obviously the cause of these drift instabilities are not understood and much more work will be required to eliminate them. Tasks are currently underway to see if these instabilities are reduced by using "cleaner" epitaxial layers. For instance, a number of impurities such as large amounts of Ca and significant amounts of Na and K were detected in the epitaxial n-type layers using SIMS analysis. Recent SIMS data show that newer n-type layers have much lower concentrations of these types of impurities. However, results using this material show no improvement in the type of instabilities described above.

CONCLUDING SUMMARY

The dielectric strength at 350°C of thin thermal oxides grown on SiC wafers can be as high as that grown on Si wafers with an absence of low field breakdown defects. However, the dielectric strength of oxides grown on SiC wafers previously subjected to an N ion-implant process is low with an occurrence of low field breakdown events. Gate oxide reliability was tested at 350°C. When positive bias is applied to the gate electrode the TDDB MTF is extremely short; however, with applied negative bias the TDDB MTF is extremely long; SiC wafers previously exposed to N ion-implantations, annealed and oxidized had very short MTF.

Methods of producing ion implanted diodes and heavily doped n^+-contact regions were developed. Both N and B ion implantation investigations were carried out. Low leakage and high reverse breakdown diodes were produced. Charge trapping was detected in N implanted n^+-p-diodes.

The studies of SiC interface characteristics using MOS capacitors and correlation with thermal characteristics of n-channel inversion mode MOSFETs made on Al doped p-type substrates showed the following: (a) the inverse temperature dependence of the effective channel mobility is due to fast interface states which correlates with the Al concentration in the substrate; and (b) correlation with SIMs data shows that Al contamination produces MOS C(V) curve hysteresis loops that indicate the presence of slow traps. In contrast, MOS devices made on N doped n-type substrates exhibits very low slow and fast trapping surface state densities and low oxide fixed charge. Also, n-channel depletion mode MOSFETs have higher channel mobilities and as expected a normal mobility temperature dependence.

Because of the above problems when using p-type material, it was decided to use n-channel depletion mode technology for fabricating SiC ICs. An operational amplifier based on depletion mode MOSFET technology was therefore designed and fabricated. Functional operation was demonstrated at 300°C. Instability drift problems were noted at 350°C and experiments are being done to further define this problem.

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